

4. Project Planning Phase

4.1 Work Breakdown Structure (WBS) Matrix

Project engineering execution tracks sequentially across clear, isolated infrastructure development blocks:

Work Block 1: Discovery & Quantitative Formulation (Milestone 1)

- 1.1 Analyze market gaps and detail core operational requirements.
- 1.2 Formulate mathematical relationship decay algorithms.

Work Block 2: Architecture Setup & Database Layout (Milestone 2)

- 2.1 Design relational schema topologies and integrate vector column definitions.
- 2.2 Configure core FastAPI routing foundations and handle error handling protocols.

Work Block 3: AI Layer Integration & Worker Pipeline (Milestone 3)

- 3.1 Connect asynchronous task processors and deploy local NLP vector tools.
- 3.2 Build modular text generation layers and automate notification flags.

Work Block 4: Validation, Testing & Operational Release (Milestone 4)

- 4.1 Execute full endpoint testing and fix core data tracking edge cases.
- 4.2 Configure container orchestration setups and verify production cloud services.

4.2 Project Schedule & Gantt Milestones

The project timeline spans 16 working weeks, mapping development workflows from initial concept research through final system deployment:

Project Task	Operational Milestone Focus	Timeframe	Dependency Verification Status
Task Phase 1	Research, Core Mathematics, and Solution Concept Design	Weeks 1 - 3	None (Project Initiation)
Task Phase 2	Database Mapping, Table Structuring, and Schema Finalization	Weeks 4 - 5	Task Phase 1 Sign-off Clear
Task Phase 3	FastAPI Framework Setup and Ingestion API Implementation	Weeks 6 - 9	Database Schema Validated
Task Phase 4	NLP Services Configuration and Async Decay Integration	Weeks 10 - 12	Core API Framework Stable
Task Phase 5	React UI Development and Interface Component Integration	Weeks 13 - 14	All Ingestion APIs Online
Task Phase 6	End-to-End Testing, Security Auditing, and Cloud Launch	Weeks 15 - 16	All Feature Modules Completed

4.3 Quantitative Risk & Failover Strategies

Potential technical challenges are evaluated below along with production mitigation protocols:

Risk Code	Risk Scenario Context	Impact	Mitigation Engineering Protocol
RSK-SYS-01	NLP Vector Generation Failures	High	Enforce strict temperature rules ($T = 0.2$) and integrate fallback validation layers.
RSK-SYS-02	Database Session Overload	Medium	Deploy Redis rate limiters to queue high-volume analytics tracking events.
RSK-SYS-03	Data Exposure Vulnerabilities	Critical	Implement database column-level encryption using secure hardware keys.